

Module 2

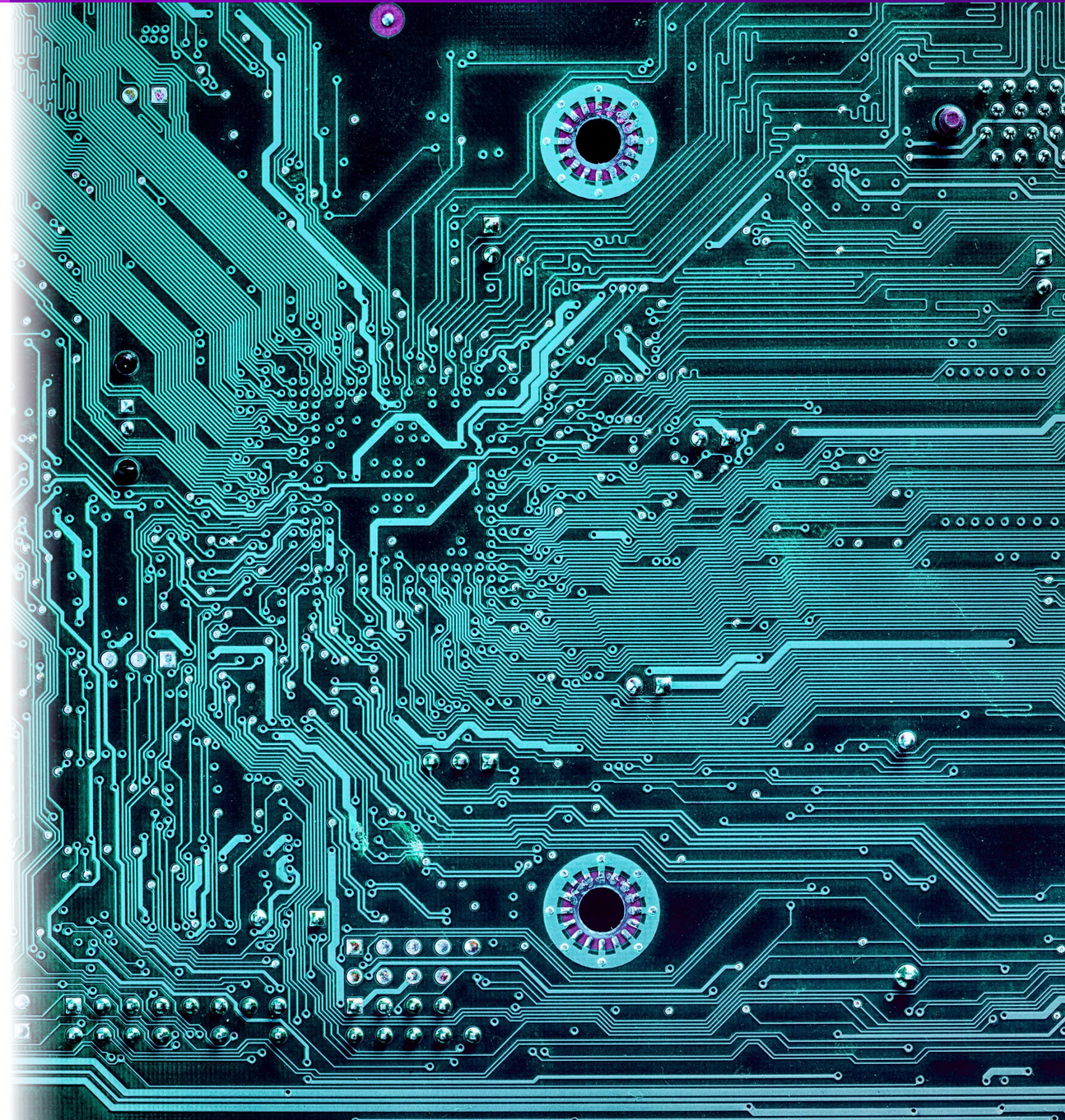
New Devices and Circuits

MTE 220 – Sensors and Instrumentation



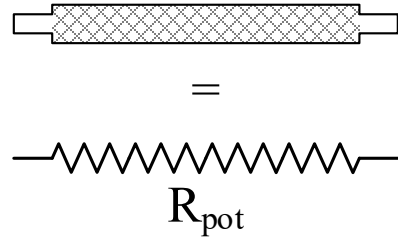
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FACULTY OF
ENGINEERING

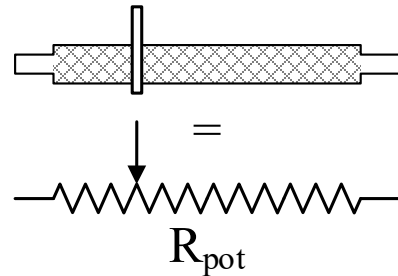


Potentiometers: Variable Resistors

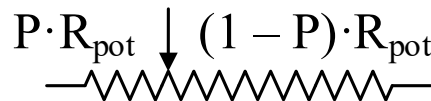
- Take a long strip of resistive material



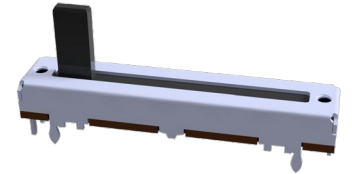
- Add a sliding “wiper”



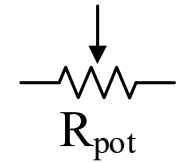
- Splits R_{pot} based on wiper position, P



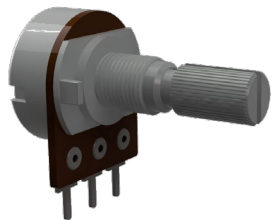
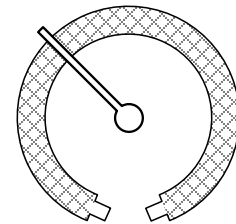
- This is a Sliding or Linear Potentiometer



- Potentiometer symbol

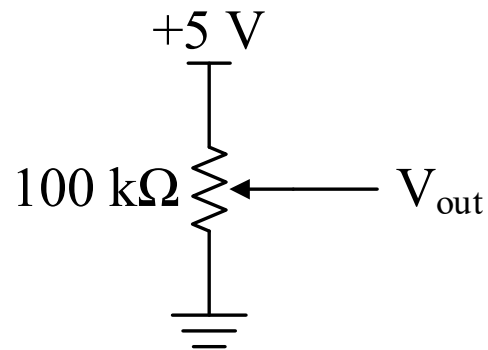
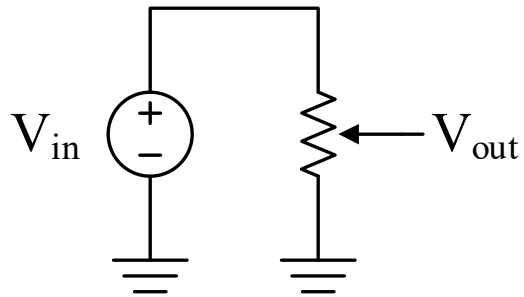


- Curve the material to make a Rotary Potentiometer

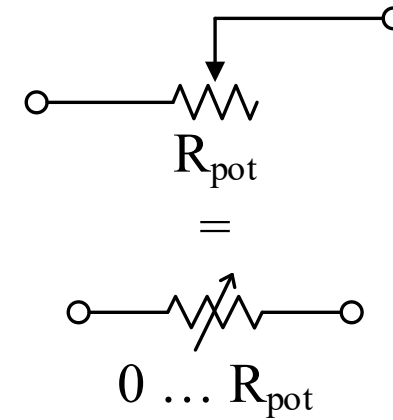


Potentiometer Use Cases

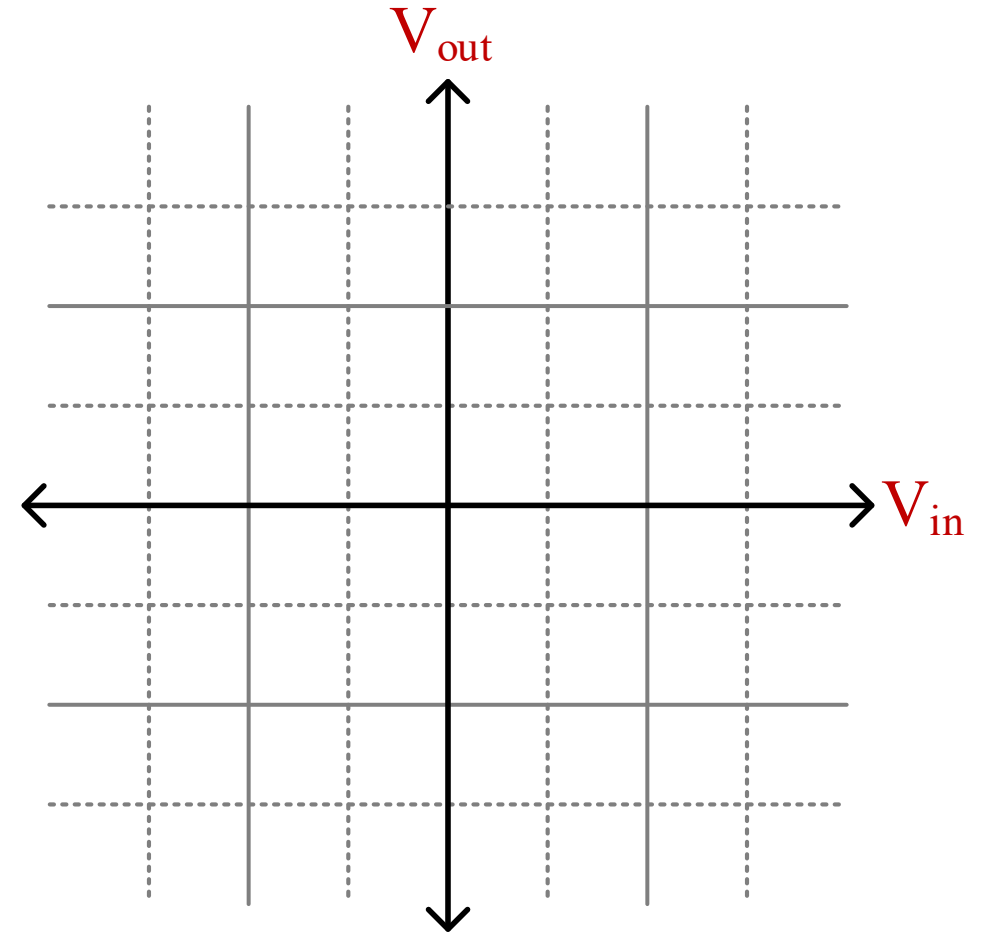
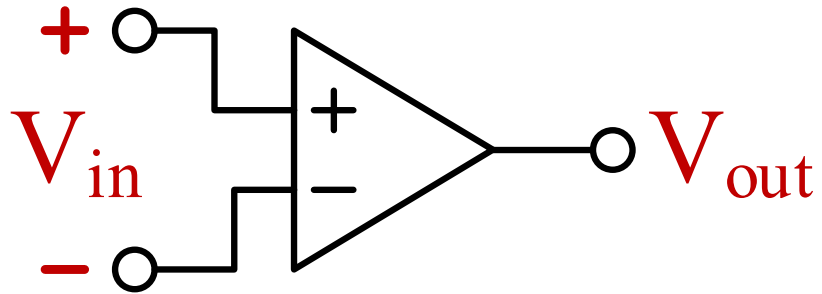
Built-in Voltage Divider



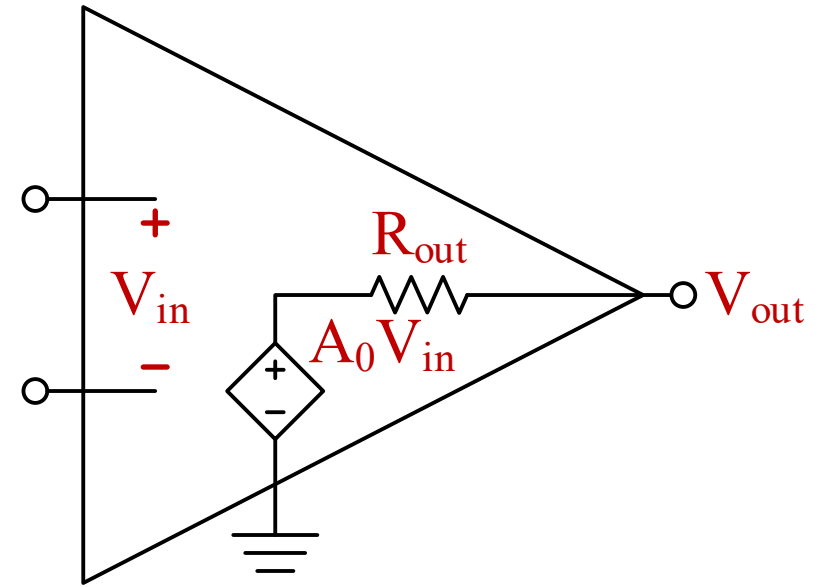
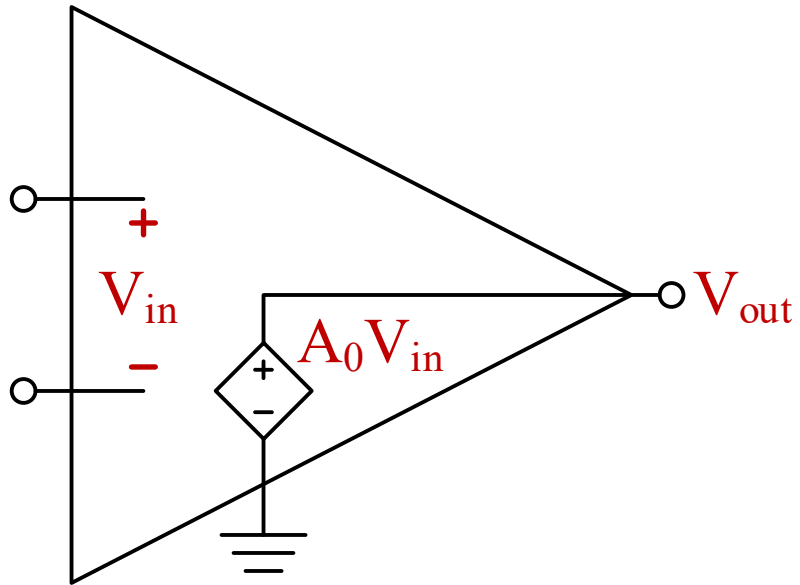
Variable Resistor



Opamp Models – Real and Ideal Behaviour

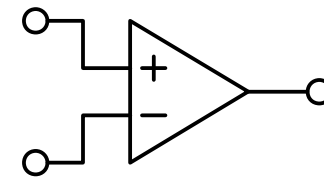
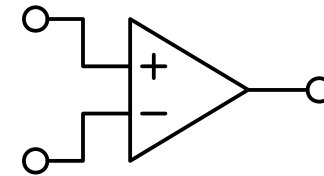
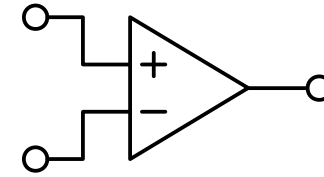
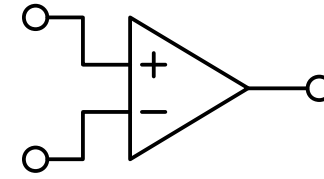


Opamp Equivalent Circuit



Feedback: Open- vs. Closed-Loop

- Open-Loop = No feedback
 - Turning a car with your eyes closed
- Closed-Loop = Feedback
 - Turning a car with your eyes open
- Negative Feedback = Minimizing a difference
 - Toilet tank stops filling once a certain level is reached
 - A fridge door closes itself to stay shut
- Positive Feedback = Maximizing a difference
 - Magnet to make door stay shut



Negative and Positive Feedback

Negative Feedback

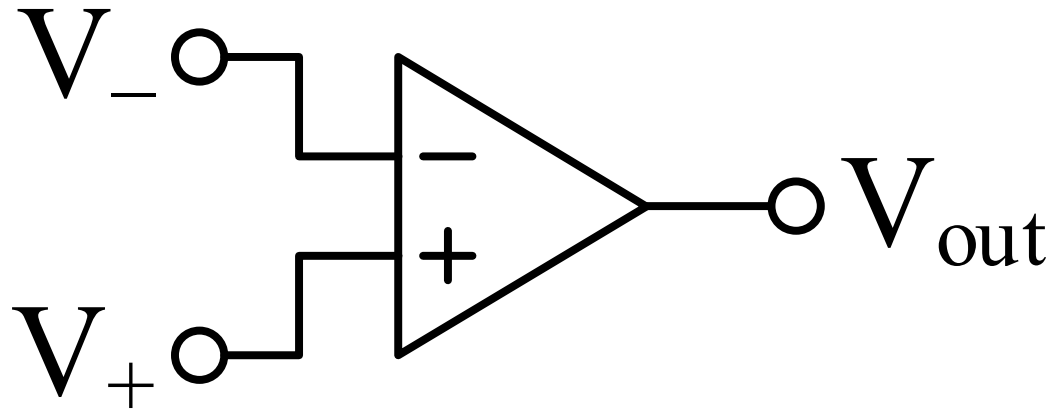
- Stabilizes a system by counteracting changes
- When a system deviates from its desired state, negative feedback works to bring it back towards that state.
- Examples
 - Thermostat in heating/cooling systems
 - Automatic cruise control in vehicles
 - Human body temperature regulation
 - Door closer spring
 - Industrial chemical process control

Positive Feedback

- Amplifies the occurrence of an event or process
- When a change occurs in a system, positive feedback reinforces that change
- Examples
 - Avalanche photodiodes
 - Microwave oven magnetron
 - Economic boom and bust cycles
 - Laser oscillators
 - Feedback in sound systems

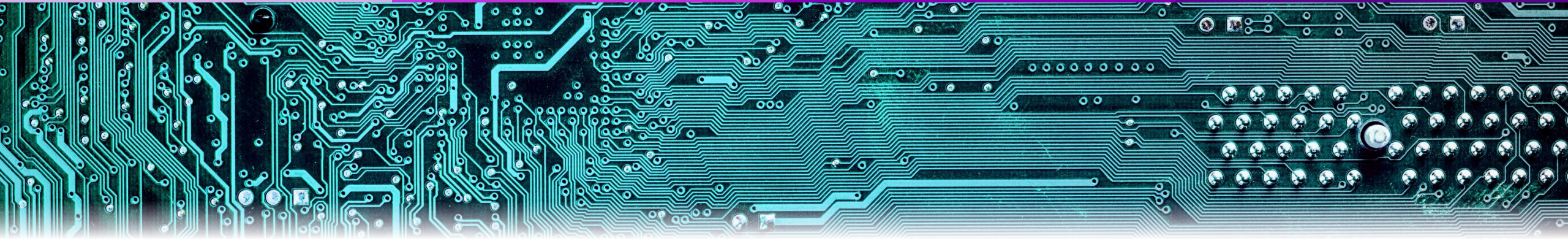


Using Opamps



- Open-loop
 - Comparator
- Closed-loop, negative feedback
 - Inverting, noninverting configurations
 - Summing, difference
 - Integrator, differentiator
 - Active filters
- Closed-loop, positive feedback
 - Schmitt Trigger
 - Oscillators

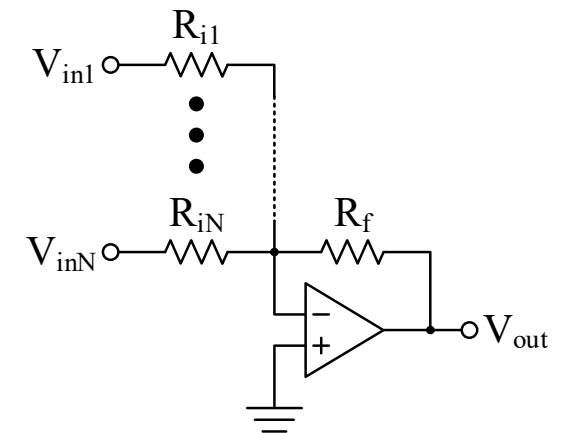
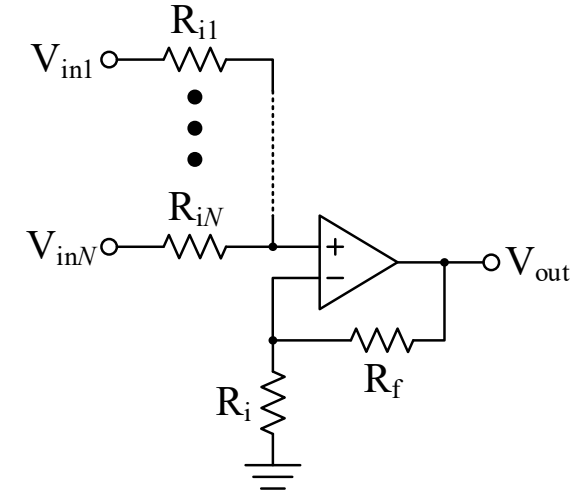
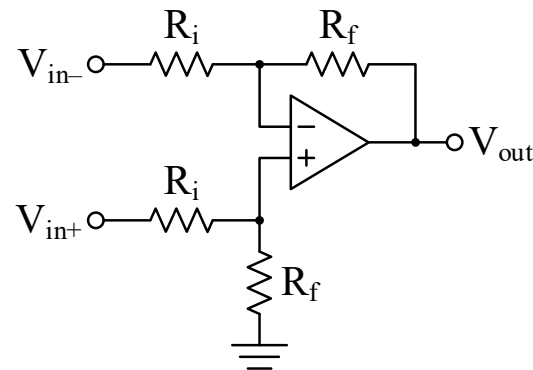
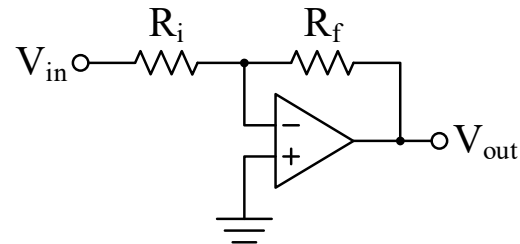
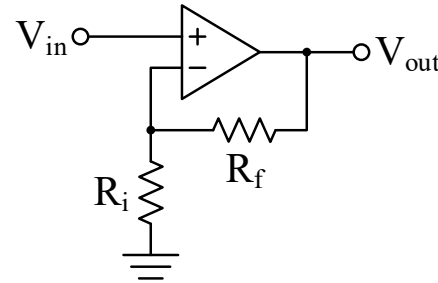




2.1 – Opamps: Negative Feedback

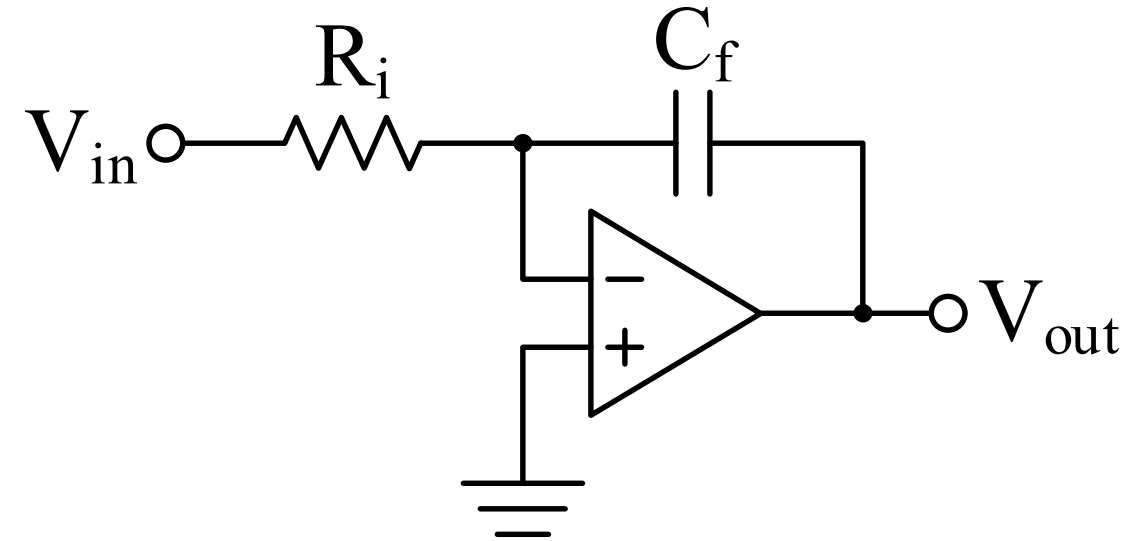
Standard Negative Feedback Configurations

- There are standard opamp circuits that achieve simple functions
- Analyzed in MTE 120 – now we use them



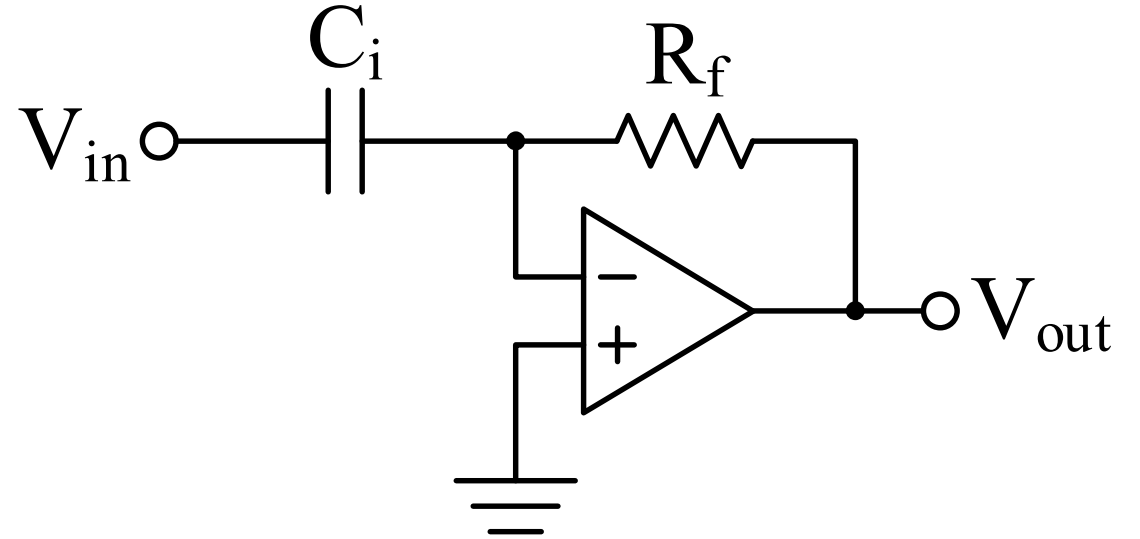
Opamp Integrator

- V_{in} controls the *slope* of V_{out}
- Useful in signal generators, error accumulators, charge measurement, and filtering



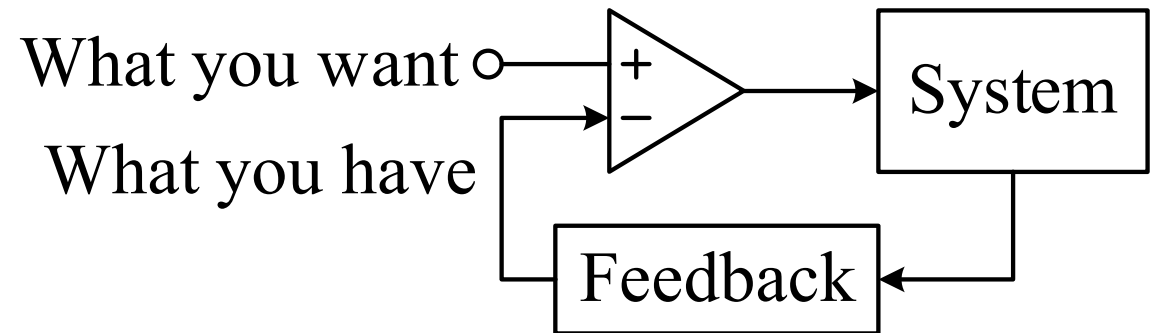
Opamp Differentiator

- The *slope* of V_{in} controls V_{out}
- Useful in edge detectors, rate sensing, transient detection, and filtering



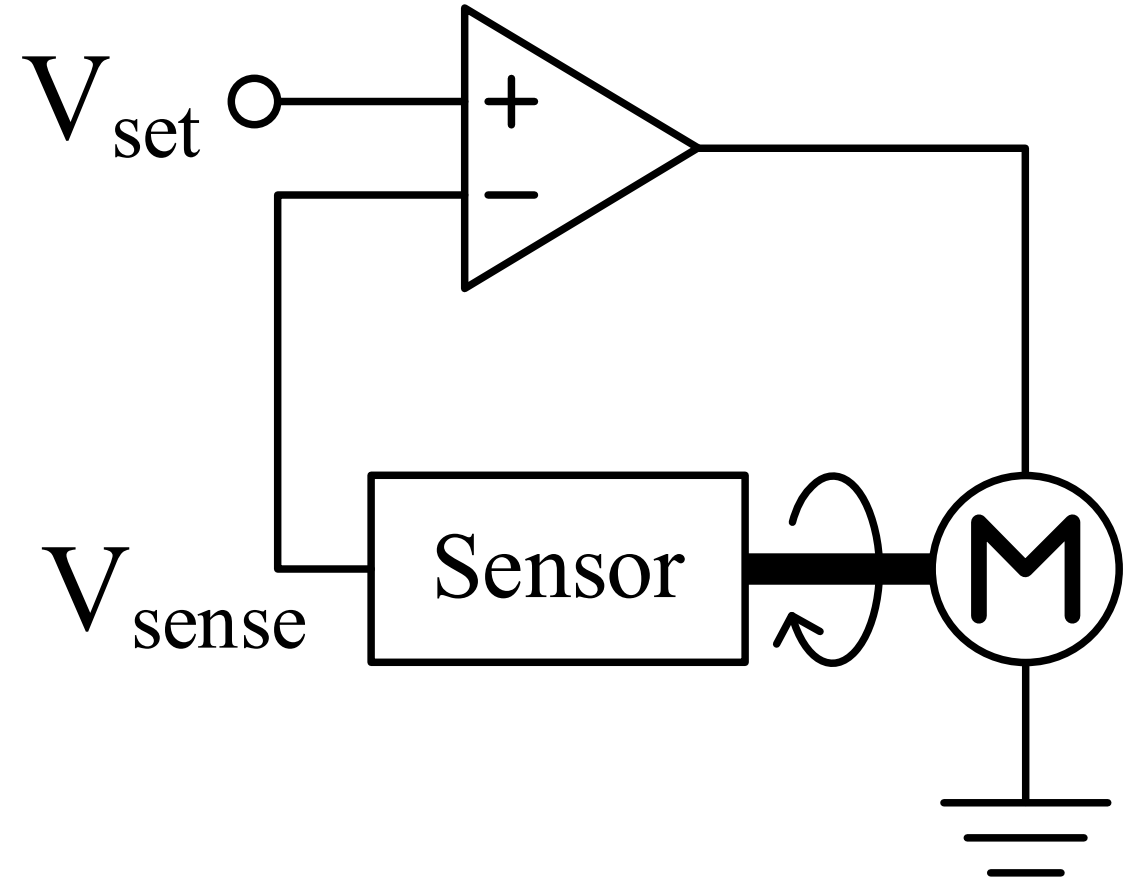
Opamp as a Negative Feedback Controller

- V_+ is what you want
 - ...to see at V_-
- V_- is what you have
 - ...measured from the somewhere you care about
- The difference between what you want and what you have is the error
 - **Error = Want – Have**
 - Opamp $V_{in} = V_+ - V_- = \text{Error}$
Opamp $V_{out} = A_o \times \text{Error}$



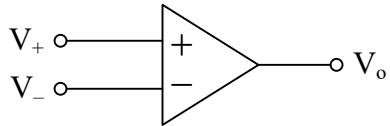
Example: Speed Controller

- The sensor outputs 0 V @ 0 rpm and 5V @ 1000 rpm
- What should V_{set} be to make the motor spin at 100 rpm?

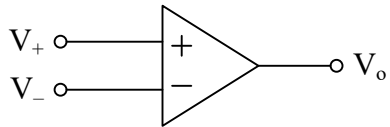


Nonideal Opamp Parameters

Open-Loop Differential Gain

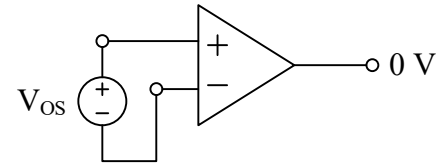


Common Mode Gain

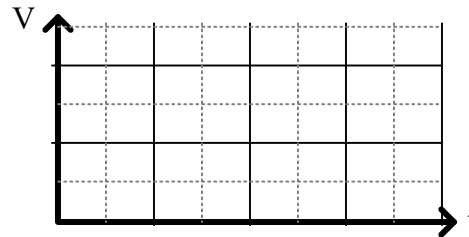


Parameter	Typical range	Ideal values
Open-loop gain, A	100k to 10M [V/V]	∞ [V/V]
Input resistance, R_i	100k to 10T Ω	$\infty \Omega$
Output resistance, R_o	0.1 to 100 Ω	0 Ω
Supply voltage, V_{DD}	1 to 24 V	∞ V

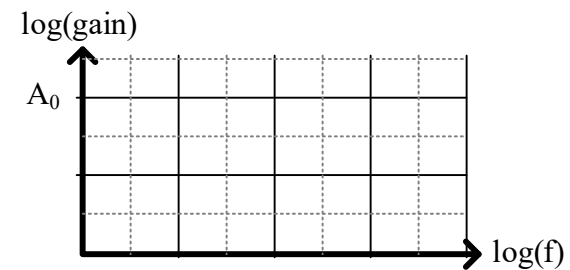
DC Offset



Slew Rate



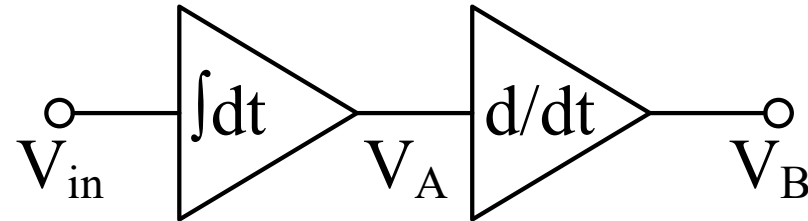
Gain Bandwidth Product



LTSpice Modeling Challenge



- Create a cascaded integrator and differentiator that cancel each other



- Mathematically, V_B should equal V_{in}
- Place $\rightarrow$ [OpAmps] $\rightarrow$ “UniversalOpAmp2”
- Create a bipolar supply as shown to give $V_{DD} = +5\text{ V}$, $V_{SS} = -5\text{ V}$
- Set both stages to have the same time constant
- What sorts of input signals “make sense” for testing your circuit?
- Notice anything strange about your differentiator’s output?

